TRACE-RPG: Auditable Neuro-Symbolic Commit Gates for Interactive Game Worlds

Anonymous Author(s)

***Abstract*—Fluent game dialogue does not establish executable, authorized state transitions. TRACE-RPG treats generated events as untrusted proposals: strict parsing, external policy, seven predicates in six families, bounded repair, and pre-commit revalidation gate mutation. Typed-error repair ρ reads only the candidate/errors, never authoritative state. Unkeyed SHA-256 links, semantic replay, and complete accounting audit outcomes. Separate evidence covers one-world offline conformance, single-model live screening, non-authoritative closed-world graph simulation, and participant-free Godot engineering. Offline evidence is mechanism-only. At $K{=}1$, guided/blind commits were 5/5 versus 0/5 only in a constructed repairable policy-blind regime. No lane establishes population efficacy, model superiority, player experience, or commercial-engine performance.**

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

INTERACTIVE generation can require absent items, assert facts unknown to the speaker, disclose forbidden quest information, introduce unauthorized effects, or regress quest stages. Executable text environments distinguish plausible utterances from valid transitions [1], [2]; grounded dialogue and personalized quests demonstrate useful quest, entity, and graph context [3], [4], not mutation authority. **TRACE-RPG treats generated events as untrusted transaction proposals; only an external deterministic commit gate permits canonical mutation** (Fig. 1).

The protocol contributes **C1—Contracts:** versioned trust-boundary parsing (Sec. IV; E1, ENG1); **C2—Controller:** checks, bounded repair, unchanged fallback, and pre-commit revalidation (Sec. IV-C; E1, ENG1); **C3—Audit:** linked semantic replay/continuity (Sec. IV-D; E1, ENG1); **C4—Harness:** assignment-complete accounting even without proposal traces (Sec. V; E1); and **C5—Repair:** typed $\rho(a, E)$ versus blind retry/state-reading oracle (Sec. IV-C; E1, E2). Tables I and IV map claims to precedents, lanes, and ceilings. Additive NPC-ENG1/NPC-SCR1 are descriptive, not C6/C-RESULT; Sec. V fixes ownership, denominators, provenance, and core Python–hard-policy/normal-play boundaries.

Lane denominators and provenance are non-interchangeable; none supports population efficacy, model ranking, player/retrieval benefit, or C6.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1, E2	frozen classes + one regime screen

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table IV. All rows are authored-fixture verified; none supports an efficacy claim.

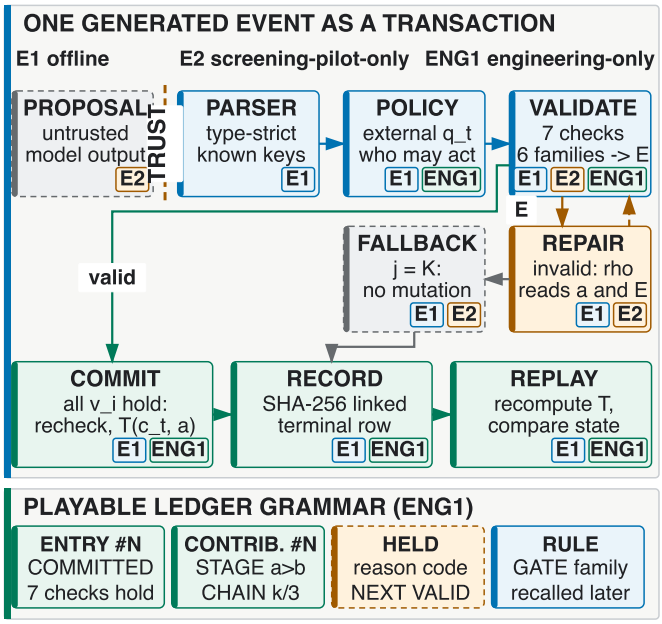


Fig. 1. Transaction boundary: external policy and c_t drive checks; typed errors feed bounded ρ (dashed) or unchanged fallback. Passing candidates are rechecked, committed, recorded, and replayed. Core tags denote E1/E2 and legacy ENG1; additive NPC `npc-goal-recovery/1.1.0` is Godot-authoritative intent/planning, not a demonstrated core Python–hard-policy round trip. The lower strip shows committed-snapshot ledger grammar.

II. Related Work and Research Gap

A. Interactive Narrative and Game Agents

KNUDGE grounds dialogue in quest/entity specifications [3]; graph quests [4], memory–reflection–planning [5], and trainable role agents [6] address grounding/believability. Player-preference/dialogue-quality studies [7], [8], [9] and a 2025 preprint's role-sensitive stability–improvisation trade-

Manuscript prepared for anonymous review.

offs [10] motivate separating hard admissibility from naturalness/player experience.

Experience-driven generation, observer-labelled engagement, and limits of vision-language engagement inference motivate future affect work [11], [12], [13]; affect is non-authoritative and absent here.

B. Environments and Benchmarks

TextWorld/ScienceWorld expose interactive state [1], [2]; MineDojo adds embodied tasks [14], and GVGAI separates agent/game/content tracks [15]. AgentBench, AgentBoard, SOTOPIA, and BALROG cover heterogeneous, process-diagnosed, social, and long-horizon settings [16], [17], [18], [19]; automated playtesting complements human evaluation [20]. We identified no reviewed protocol combining transactional authorization, typed repair, semantic replay, and complete accounting; this is not a first-ever claim. NCP-Bench tests narrative commitment/fact consistency, not TRACE-RPG or player outcomes [21].

C. Constraints, Neuro-Symbolic Validation, and Retrieval

Structural constraints [22], [23], [24] differ from state-relative authorization. LLM-Modulo uses external sound critics [25]; SayCan checks skill affordances [26]; AgentSpec enforces trigger–predicate rules [27]. Setting the DC checks tool preconditions [28], whereas we parse proposals against independent policy. STORY2GAME generates its own preconditions/effects [29]; PANGeA uses LLM rule judgement/refinement [30]; IVIE symbolically validates incremental worlds [31], and its successor restricts LLMs to authored transformations [32]. These precedents do not resolve our transaction/audit/repair questions or imply superiority beyond encoded predicates. JSONSchemaBench's structural compliance/coverage/quality measures do not establish semantic authorization [33].

G-Retriever, HippoRAG, and Mem0 address subgraph retrieval/long-term memory [34], [35], [36]; Generative Agents and Character-LLM address behavioral memory/persona [5], [6]. Such context has no commit authority here; benefits are unmeasured.

D. Auditability and Reproducibility

Process metrics, sparse-run caution, reproducibility, and environmental reporting motivate failure retention and frozen artifacts [17], [37], [38], [39]. Human/LLM evaluation and future participant analyses require construct/rater/bias controls, justified margins, and random-effects structures [40], [41], [42], [43], [44], [45]. Record-and-replay supports experience reuse [46]; TRACE-RPG claims only semantic replay of admitted transitions. Leakage checks protect frozen denominators but do not create efficacy evidence [47].

The contribution is the C1–C5 combination, not invention of enforcement [27], game-state management [28], [29], feedback repair [48], replay [46], action interposition [49], preconditions/effects [50], transition enforcement [51], abort [52], social logic [53], or belief-aware planning [54]. Reporting omissions do not establish implementation absence.

III. Problem Definition and Guarantee Boundary

Validation input at step t is $c_t = (G_t, q_t)$: typed snapshot fields G_t and externally authored action-policy, disclosure, and quest-stage constraints q_t stored with them. Terminal history $h_{<t}$ is separate from `WorldState`; predicates never read it, nor is state reconstructed from logs. An accepted one-step transition updates G_t under unchanged q_t; policy-version rotation is out of scope. Retrieval, narrative scores, model rationales, memory summaries, and future affect estimates are non-authoritative proposal context.

An untrusted proposer emits candidate a_t. Strict parsing first enforces the shared candidate-key contract, including rejection of unknown top-level fields, and establishes the declared schema. The validator returns

$$V(c_t, a_t) = (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}, E_t), \tag{1}$$

where E_t is a structured error set. Mutation is

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7} v_i = 1, \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

Deterministic valid/invalid/repair/fallback/replay/fault fixtures test four implementation invariants, not general theorems. **I1:** exhausted validation or pre-commit proposal/controller failure returns the prior state. **I2:** returning callbacks and $K \geq 0$ imply at most $K + 1$ recorded attempts plus a successful commit's defensive validation; without enforced callback deadlines, this is not wall-clock termination. **I3:** accepted `CandidateAction.effects` lie within `ActionPolicy.allowed_effects`, and policy explicitly authorizes quest-stage targets. **I4:** frozen schemas/deterministic transitions let accepted traces reconstruct their stated terminal state.

These invariants assume correct policy authorship, faithful extraction into declared candidate fields, deterministic software versions, returning callbacks, and single-process file ownership. Replay revalidates the recorded structured candidate and transition; it does *not* verify the provenance, prompts, or stochastic generation of intermediate repairs, and natural-language implications omitted from structured fields remain outside the guarantee.

IV. TRACE-RPG Architecture and Algorithms

A. Trust Boundary and Contracts

The model/recorded adapter lies outside the canonical owner (Fig. 1). Parsing rejects ambiguous numeric/collection types and unknown top-level fields; separate action policies define required preconditions/allowed effects. The versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits; only valid commits authorize mutation. Interface separation demonstrates neither commercial-engine portability nor proposal-default/replay-parser equivalence: evidence supports unknown-key rejection parity only. Equation (1) has seven deterministic checks (Python Boolean predicates); the Godot dashboard groups them into six state-relative families (quest eligibility/stage share one). NPC-DISC-1's seven DISC rules are a separate disclosure contract, not either count.

TABLE II
ENCODED VALIDATOR BOUNDARY

Predicate	Rejects	On failure
Action policy	unknown action/type	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	undeclared known fact	unchanged
Disclosure	forbidden fact	unchanged
Quest eligibility	ineligible quest action	unchanged
Quest stage	regressive or unauthorized stage	unchanged

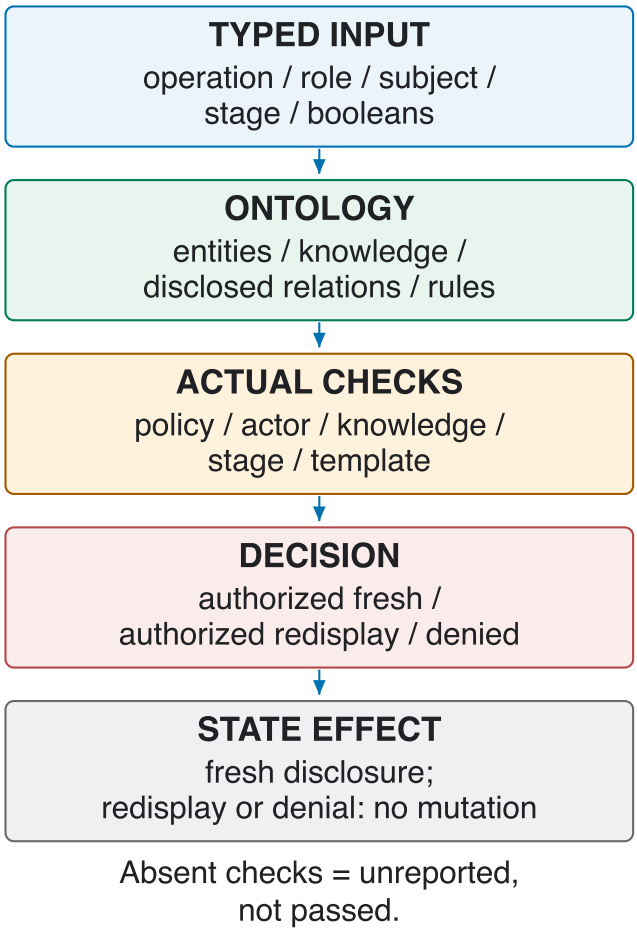


Fig. 2. Content-first presentation contract: typed input maps to ontology entities/relations and authored proposal/authorization checks; the observed decision is allowed or denied, and only allowed transitions change canonical state. Runtime policy and authored methods ontology remain distinct; this is neither live capture nor a hidden-state claim.

B. Fixed-goal NPC integration

The NPC lane uses `npc-goal-recovery/1.1.0`: Codex proposes typed intent, Python verifies public claims and computes a bounded public-projection plan, and Godot alone authorizes and writes canonical state. The fixed goal is harbor-signal restoration followed by the authorized tide lead for `captain_mira`; utterances and model output cannot replace it. Godot checks session/epoch/request/scenario/NPC/goal/version, policy/config hashes, revision/state hash, deadline/budget, and issued-action binding before writing. The service is loopback-only at `127.0.0.1:8768`; browser origins are `localhost` and `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1 is a declaration-derived closed `reveal_hint` contract: 48 normal, 4 boundary, and 8 malformed cases. Generated results and Tables IX–XI bind packets, canonical IDs, approved templates, arm observations, and fixtures. Typed input maps to ontology entities/authored checks; Godot alone rechecks service bindings and mutates canonical state. Outcomes show bounded contract consistency, not ontology superiority or efficacy. These 60 disclosure cases are separate from the original NPC `verified-run-006` assignments; they neither rerun nor enlarge that historical denominator.

1: $a \leftarrow \textsc{Propose}(c)$; $j \leftarrow 0$
2: **repeat**
3: $(ok, E) \leftarrow V(c, a)$; $\textsc{Record}(a, E)$
4: **if** ok **then**
5: **assert** $V(c, a).ok$; **return** $\textsc{Commit}(c, a)$
6: **if** $j = K$ **then return** $\textsc{Fallback}(c)$
7: $a \leftarrow \textsc{Repair}(c, a, E)$; $j \leftarrow j + 1$
8: **until terminal**

Fig. 3. Validate–repair–commit procedure. At most $K+1$ candidate attempts are recorded; the successful path performs a defensive validation before mutation.

C. Validate–Repair–Commit

Fig. 3 gives dependency-free pseudocode: "Record" appends in memory; persistence follows terminal consistency checks.

Repair cannot mutate state; exhaustion returns unchanged fallback. The typed-error operator (`counterexample_guided_repair`) reads only candidate a and errors E (its sole state-derived input), never authoritative `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

Table III defines deterministic edits $\delta(e)$; $\oplus$ applies them. At most one field changes per error entity, hence at most $|E|$ fields per attempt. The fixed-point property holds only for identical E, not changing errors or convergence. Conflicting requirements/rejections on one field are irreparable and untouched. The state-reading reference ignores errors and reconstructs preconditions/effects/both quest-stage fields from authoritative state. It never edits required objects/used facts/disclosed facts: deleting these declarations would launder violations, not repair them. This oracle is not deployable-repair evidence.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 row checksums and proposal–trace links detect accidental/tested changes, not signatures, writer identity, or joint content/checksum rewriting. Replay strictly parses prior state/candidate/validation pairs, requires invalid nonfinal attempts, recomputes the terminal transition, compares final state, and checks adjacent JSONL continuity. It never reruns repair generation. Request→decision→result order and

TABLE III
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	Edit $\delta(e)$ on the candidate
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate-local edit would launder or needs state)

TABLE IV
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	Design	Unit / budget	Comparison	Ceiling
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E2	Screening-pilot-only RQ2: one hosted proposer, matched candidate	5 cells $\times$ 5 calls; $K{=}1$	ρ vs blind per regime	screening-pilot-only transfer screen
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 smoke; 5 archetypes	authored fixture paths	engineering-only conformance

Lanes are non-interchangeable with different denominators; E2 is screening-pilot-only and ENG1 is engineering-only. None establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

first-issued-action/exactly-one-receipt binding imply neither duplicate-action-ID issuance prohibition nor cryptographic authentication.

Terminal classes are commit, symbolic fallback, adapter failure, and controller failure. The latter preserves state/assignment even without a completed candidate trace; complete accounting need not mean a trace per row. Pre-execution keys bind run/arm/scenario/seed/model/revision and controller-configuration/input/prior-state hashes. Aggregation rejects duplicate/missing/unexpected keys and separates symbolic/adapter/controller failures.

V. EVIDENCE LANES AND BOUNDARIES

Table IV fixes lane designs, units, comparisons, and inference ceilings.

A. *E1 Offline Conformance Design*

E1 tests (PQ1) valid commit/invalid-policy immutability; (PQ2) frozen class outcomes for rejection-only, blind unchanged retry, ρ, and state-reading; (PQ3) designated rejection of checksum/state-semantic/linkage/continuity mutations and pair-write rollback after injected second-file append failure; (PQ4) strict adapter/manifest retention of one classified row per assignment or fail-closed rejection; and (PQ5) unknown-top-level-key rejection by proposal and replay parsers.

Purposive conformance fixtures cover validator/state isolation, repair arms, record/trace fault injection, and adapter/accounting contracts; they are not population samples. Before execution, each repair fixture fixes its class and arm expectations: *guided-repairable* (error payload suffices for ρ), *oracle-only* (requires authoritative state), or *irreparable* (no arm may commit). Denominators count unique cases: 13 gate fixtures (one valid control, 12 isolated negatives) and 12 separate repair cases.

Measures are exact conformance counts: validator agreement, state-preserving rejection, repair-arm outcomes by class, designated fault rejection, replay, manifest violations, and unknown-key proposal/replay parity. They are not detector attribution, p-values, or population intervals. Adapter latency/token constants exercise accounting propagation only; the frozen packet binds fixtures, assignments, schemas, results, and code revisions.

B. *E2 Live RQ2 Screening*

E2 compares blind unchanged retry with ρ at $K{=}1$: five seed-indexed hosted `gpt-5.6-sol` calls per cell provide one candidate to both arms. Policy visibility and zero-effect-route availability vary. Uncontrolled seeds, unpinned revision, unavailable token accounting, and constructed nonpopulation states limit reporting to exact counts/failure classes, without p-values, intervals, or model ranking.

C. *NPC-ENG1 Fixed-goal engineering lane*

NPC-ENG1's 16 deterministic native rule-control cases cover bootstrap, identity/hash binding, intent/plan/role/result transitions, negative gates, stale/duplicate/deadline paths, and replay. Stable v5 defines 16 engineering assignments/12 screening recipes; verified-run-002, verified-run-003, and verified-run-004 remain immutable historical packets. Historical preflight is outside NPC-SCR1. Only the validated `research/academic-pipeline/npc-goal-recovery/verified-run-006/` packet and repository runner/validator support final status/count/latency and reproduction: protocol/engine evidence only.

D. *NPC-SCR1 Descriptive screening*

NPC-SCR1 fixes 12 unique cases $\times$ two repetitions $\times$ three arms (P/R/O): 72 assigned rows, 24 per arm. NPC-ENG1 contributes 16 engineering definitions, giving 88 assignments over 28 definitions ($R/P/O = 40/24/24$), separate from legacy ENG1/E1. All one-goal failure strata remain assigned, with recovery-eligible and unreachable rows separate. P is authenticated Codex OAuth without fallback; requested/returned model labels and provider revision are unavailable. R is authored rule/grammar; O is a protocol-declared same-family auxiliary with family match unverified and no independent gold (auxiliary-source fraction 100%). Historical packets remain provenance only. Observed outcomes, assignment-failure splits, stale-ACK controls, and epoch-local replay are reported in the generated NPC results/tables as descriptive protocol evidence, not efficacy, semantic gold, accuracy ceiling, model ranking, or C6 evidence.

TABLE V
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	Info source	Commits / cases			Mean edits[e]
		G-rep.[f]	O-only[f]	Irrep.[f]	Fields
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e]Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f]Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VI
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	Exact outcome	Interpretation
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

E. *NPC-DISC-1 Closed-world disclosure screening*

NPC-DISC-1 evaluates only `reveal_hint` for the sealed-lighthouse scenario; its frozen strata, arm semantics, and exact observations are specified once in Sec. IV-B1, the generated disclosure results, and Tables IX–XI.

F. *E3 KG/Ontology Link-Proposal Simulation*

The third lane is an engineering-only closed-world simulation, separate from the authoritative `WorldState`: 43 graph nodes and 106 reference edges, 24 reviewed typed edges, passing type/domain/range checks and six competency questions, with six graph-edge-withheld authored targets used both to select and summarize the retained strategy—not an untouched or external test set—and scored under seven fixed strategies. The retained typed-lexical strategy (S2) recovered 6/6 authored targets ($P = R = F_1 = 1.000$, realistic-tie $\mathrm{MRR} = 0.944$, Brier $= 0.131$, $\mathrm{Sem@3} = 1.000$); the degree baseline recovered none. These are construction results: not OWL/SHACL conformance, semantic completeness, runtime retrieval, or evidence of any efficacy claim.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The pilot loader admits a fixture set only when it isolates every implemented validator code exactly once alongside one valid control, so the 13/13 agreement and the observation of all 12 implemented error codes are *construction invariants of the harness*, not measured success rates. What the run adds is the part the loader does not enforce: each of the 12 isolated negative fixtures showed observed agreement with its authored expected code rather than reaching a different one, and every noncommit path preserved the complete canonical state. This is implementation conformance to an authored structured-field oracle over a single world state, not accuracy against independent semantic labels.

B. Repair-Arm Comparison on Designed Fixtures

The 12 initially invalid designed cases are partitioned into frozen repairability classes: 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. The validator-error-guided operator ρ committed 5/5 guided-repairable cases and, by design, 0/1 oracle-only and 0/6 irreparable cases; every ρ commit edited a mean of 1.0 candidate fields. ρ reads only the prior candidate and typed validator errors E; those errors are its sole state-derived input, and every failed arm execution left the canonical state unchanged. The state-reading reference callback committed 5/5 guided-repairable and 1/1 oracle-only cases, with 0/6 irreparable. These exact counts separate the three mechanisms on the designed fixtures: blind retry never converts a validator error into a commit, ρ converts exactly the cases whose error payload carries sufficient information, and the oracle additionally converts the case requiring state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION`+`UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each became valid through the corresponding Table III edit $\delta(e)$ alone. This is operator conformance on frozen fixtures, not repair quality of any live model.

C. Integrity, Boundaries, and Assignment Accounting

All 10 faults prespecified as detectable were rejected by their designated check operations (10/10). Because this pilot does not compare stable typed detector codes, it does not establish detector-layer attribution. Separately, the known provenance-boundary fixture substituted a different invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay as expected (1/1 undetected). Thus, the mechanism does not protect against a writer who can recompute unkeyed checksums, and replay does not authenticate the repair-generation operation. Two open boundary sentinels—unextracted narrative disclosure and simultaneous omission of a required object from candidate and policy—were intentionally accepted at the encoded layer (2/2), with

TABLE VII
E2 SCREENING-PILOT-ONLY: LIVE SCREENING EXACT COUNTS (5 CELLS, 5 PROPOSALS PER CELL, K=1)

Base / condition	Initial valid	ρ vs blind commits	Δ	Initial errors
Frozen / visible	5/5	5/5 vs 5/5	+0	—
Frozen / blind	5/5	5/5 vs 5/5	+0	—
Frozen / goal-blind	0/5	0/5 vs 0/5	+0	QSR
Signal-v2 / visible	5/5	5/5 vs 5/5	+0	—
Signal-v2 / blind	0/5	**5/5 vs 0/5**	**+5**	PEO+PEV+PPO

QSR = quest-stage regression, PEO = policy-effect omission, PEV = policy-effect violation, PPO = policy-precondition omission. Δ is ρ commits minus blind commits for matched candidates. There are 5 cells with 5 calls each: 3 cells were initially valid (5/5), while the goal-blind cell had QSR in 5/5 and 0/5 and 0/5 commits in both arms (null regime). Only Signal-v2/blind separated (ρ 5/5 vs blind 0/5). All counts are screening-pilot-only, with no inferential statistics or model ranking.

TABLE VIII
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	Assigned	Complete/fail	Goal reached	Recovery E/U
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

0 labelled as safety passes. The former unknown-top-level-field acceptance boundary was later reclassified as a closed negative regression: both proposal and replay parsers specifically rejected a complete 12-field candidate carrying one unknown key (1/1). This establishes parser rejection parity, not semantic safety. Among seven adapter/accounting assignments, outcomes were 1 commit, 1 symbolic fallback, and 5 adapter failures. Synthetic telemetry fields were populated and propagated for 2/7 assignments; these are schema-pinned constants verifying accounting-field propagation, not measurements. All 3/3 injected duplicate or missing-assignment guards failed closed. These are raw counts from synthetic offline fixtures.

D. Live Screening Pilot (Pilot-Only)

The pilot-only counts in Table VII show that Signal-v2/policy-blind had 5/5 invalid initial candidates; ρ committed 5/5 and blind retry 0/5. The other four current cells were null, while all 15/15 noncommit outcomes preserved the prior-state hash. In the superseded v1 diagnostic, up to 3 co-occurring errors were reduced to one irreparable knowledge error but still did not commit.

E. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira's fixed goal is to deliver the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent and public observations. Acquire/install remain player actions; the NPC may only commit the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await. Results and new observations determine the next plan, unlike single-candidate validator-error repair.

NPC-ENG1 comprises 16 deterministic R-arm engineering cases initialized from actual engine snapshots (assigned_failure=2, completed=14). NPC-SCR1 retains all 72 assignments and failures, 24 per P/R/O arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures; these are not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter propagates the configured selector, not a provider revision; the provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table VIII reports bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates. Efficacy remains pending; tokens and cost are unmeasured. A read-only CLI and isolated working directory were used, but tool-free inference and adversarial confidentiality were not established. This generation revalidates immutable verified-run-006 against archived source bytes dated 2026-10-17 (revision fa9fb8a). It is a historical packet result and does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. These rows remain assigned failures in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception is restricted to an adjacent leading observation request/decision pair in a positive-epoch snapshot. Epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

F. Frozen `reveal_hint` contract and disclosure evidence

This artifact consumes only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four actual result files. The packet SHA-256 is `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028`, checked directly against its sidecar. The declared normal stratum has 48 cases from `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary has 4 cases and malformed inputs have 8 cases. Boundary and malformed inputs are excluded from the normal denominator, and each trace expectation is

matched back to the packet by case ID rather than copied from a gate output.

The content-first explanation reads typed input (operation, actor, fact, stage, knowledge, and prior disclosure) into ontology entities/relations, applies authored rules, records actual checks, then reports an allowed/denied decision and its canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once. Denials and redisplays preserve canonical state. The presentation is explanatory: it does not recover hidden facts or natural-language implications and does not claim live capture.

The runtime declaration is `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, with provenance `frozen_expectation_packet`. Scope contains four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and the eligible actor `captain_mira`. Rule IDs are DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; the only approved templates are `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

The Python evaluator materializes the same cases and expectation object in three isolated arms and records semantic outcomes, check traces, errors, before/after state SHA-256 values, rendered template IDs, and replay equality. Godot is the separate game-execution boundary that owns the existing `reveal_hint` writer and canonical mutation; this producer neither reconstructs nor replaces Godot state. The mapping is traceable through operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source. 3 source hash(es) differ from the frozen baseline (game-track/godot/scripts/game3d/game_3d.gd, game-track/godot/scripts/sealed_lighthouse_machine.gd, llm-wiki/wiki/reports/2026-10-22-ontology-signals-interview.md); the packet hash is preserved and the difference is disclosed.

Table X reports the three arms over the same 60 cases with separate normal, boundary, and malformed denominators. Violations in unchecked are intentional isolated observations; general-rule and ontology-gate are separately checked for 0 frozen-expectation mismatches. Figure 4 distinguishes expected-match/replay case counts (E/R), check-record PASS/FAIL counts (C), and counts of cases with no reported checks (NR). NR does not mean skipped, pass, or out-of-scope. Table XI maps two rule-change fixtures and four paired cases to disposable policy clones; the shipped scenario is not modified.

Claims are limited to bounded engineering conformance. This evidence does not establish human gold labels, an access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. Exact input paths, hashes, and reproduction commands are bound by `paper/latex/generated/disclosure_producer_manifest.json`. Live disclosure sequence/`t_ms`, public-safe request/state hashes, and categorical checks form an engine timeline, not provider receipt time/latency or the fixed-case table. Missing/out-of-scope cells are not PASS or SKIPPED.

EXTERNAL TRACE COUNTS

	N normal	B boundary	M malformed
U	E 0/48 n=48 R 48/0 C 0/0 NR 48	E 0/4 n=4 R 4/0 C 0/0 NR 4	E 0/8 n=8 R 8/0 C 0/0 NR 8
G	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0
O	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 trace counts across three arms and normal/boundary/malformed strata. E/R count expected-match/replay cases; C counts policy-check PASS/FAIL records; NR counts cases without reported checks, not skipped checks. These are packet-bound engineering counts, not latent-state, free-language-leakage, ontology-superiority, or player-benefit measures.

TABLE IX
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	Value
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

ENG1. (Legacy; no live transport.) Godot 4.7.1 replays authored disclosure rejection/commit to the prior/offline terminal hashes. Captures are presentation-only, not evidence. Committed facts populate board/hints/coach/chain prompt; the dashboard distinguishes checks/holds/unreached/out-of-scope. Core 52/52 and four extension checks show engine-local conformance, not live model/Python transport, participant, narrative, usability, or efficacy evidence.

VII. DISCUSSION

A. Established Scope

Mechanism fixtures test state isolation, repair/fallback, corruption rejection, and row classification, not detector attribution. NPC counts (Sec. V) imply no efficacy, superiority, generalization, human benefit, cost equality, semantic gold, or accuracy ceiling.

State-reading handles oracle-only cases; ρ matches guided-repairable cases only. Live null regimes, unpinned revision, and quasi-replication preclude sample-efficiency/model-capability claims. Unlike semantic repair, ρ consumes deterministic pre-execution errors with failed arms isolated [48], [55], [56], [57]. **Disclosure interpretation.** `general_rule`/`ontology_gate` parity is frozen-case consistency, not ontology superiority/completeness/necessity, latent-state inference, or arbitrary-prose leakage freedom. Rule-change fixtures diagnose edit scope/missed paired cases.

B. Relationship to Prior Work

Planned, not executed: preregistered multi-model/oracle/blind-retry comparisons, matched-validity retrieval/memory ablations, and blinded matched-validity human study with justified power/mixed-effects; screening substitutes for none [40], [37], [45].

VIII. THREATS, ETHICS, AND ARTIFACT SCOPE

Nine boundaries apply: **L1:** encoded fields miss semantics and narrative quality is unmeasured [58]; **L2:** one world/live state gives no cross-world, cross-policy, or scale generalization; **L3:** state-reading is an oracle and classes/regimes do not establish repair quality or sample efficiency; **L4:** one hosted proposer has unpinned revision/seeds and no token accounting, while E2 has no Python–Godot authorization round trip; **L5:** no participant/affect or comprehension/agency/enjoyment/immersion/benefit evidence; **L6:** unkeyed hashes provide integrity, not authentication or tamper resistance; **L7:** concurrency is unverified under single-writer consistency; **L8:** one startup-dominated Godot fixture has samples above 16.7 ms, so portability/performance are untested; **L9:** selected fixtures/cells support no population, safety-rate, or model-ranking inference. Internal validity requires independent policy expectations/no leakage; reproducibility requires pinned software/artifacts. NPC-DISC-1 covers only frozen IDs/templates/state axes, not free-form/implicit, hidden-payload, or all-UI-channel leakage. Declaration-derived expectations are neither human gold nor holdout; Jev has no label/commit authority. NPC deadline/duplicate/stale/late-drop/budget/binding checks are protocol, not latency/quality evidence. Cancellation is local-only; `core.cancel` is unit/controller-only, not human/cross-world/independent-gold/C6 evidence. No participants/personal telemetry were collected. Future human/affect work requires ethics review, consent, compensation disclosure, minimization, retention/deletion, opt-out, and calibrated/bias-controlled auxiliary judges [40], [41], [42], [43].

IX. CONCLUSION

TRACE-RPG gates mutation and audits linked replay. Fixtures show conformance; pilot screening separates ρ/blind retry (5/5 vs. 0/5) only in the constructed repairable regime. NPC-DISC-1 adds closed `reveal_hint`/three-arm/finite-strata/redisplay-no-mutation evidence, not gold or efficacy. Cross-model/player/retrieval/memory/affect/commercial benefits remain untested; Sec. V separates additive NPC, historical `native012`, and legacy ENG1.

DISCLOSURE OF AI USAGE

LLMs assisted prose, code/tests, and citation checks; a hosted model supplied labelled screening candidates. Counts derive from deterministic runners/hash-bound receipts. Curated AI-generated UI art (portrait, seals, textures, board, icons, guide) and desk mesh have per-file provenance; engine-rendered snapshot values/hashes/ledger lines are presentation-only. Authors checked source artifacts and accept responsibility. Additive NPC inputs/tables are separate from NPC-GOAL-RECOVERY, retain all assignments, bind project/protocol/schema hashes, and mark unavailable token/cost/model-revision values unknown.

DATA AND CODE AVAILABILITY

Implementation/build sources, frozen packets, the Godot slice, deterministic evaluation, and the machine-checked contribution, reference-topic, and experiment matrices behind Tables I and IV are included. `paper/latex/generated/disclosure_producer_manifest.json` binds the active `research/npc-disclosure/results-signal-explainer-20261023-r5/` NPC-DISC-1 files to the frozen expectation packet

TABLE X
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XI
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

and sidecar and to reproducible bilingual inputs and `fig_disclosure_signals.pdf`. Earlier directories are historical receipts, not current-runtime proof; no reviewer archive or DOI deposit is claimed.

REFERENCES

[1] M.-A. Côté, Á. Kádár, X. Yuan *et al.*, "TextWorld: A learning environment for text-based games," in *Computer Games*, ser. Communications in Computer and Information Science. Springer International Publishing, 2019, vol. 1017, pp. 41–75.

[2] R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu, "ScienceWorld: Is your agent smarter than a 5th grader?" in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2022, pp. 11 279–11 298.

[3] N. Weir, R. Thomas, R. d'Amore, K. Hill, B. Van Durme, and H. Jhamtani, "Ontologically faithful generation of non-player character dialogues," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 9212–9242.

[4] T. Ashby, B. K. Webb, G. Knapp, J. Searle, and N. Fulda, "Personalized quest and dialogue generation in role-playing games: A knowledge graph- and language model-based approach," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 2023, pp. 1–20.

[5] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 2023, pp. 1–22.

[6] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-LLM: A trainable agent for role-playing," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 13 153–13 187.

[7] N. Akoury, Q. Yang, and M. Iyyer, "A framework for exploring player perceptions of LLM-generated dialogue in commercial video games," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 2023, pp. 2295–2311.

[8] M. Hochreiter, S. Krigstein, and G. Wallner, "Beyond pre-defined scripts: Player perceptions on generative non-player character dialogues," in *Proceedings of the 31st International Conference on Intelligent User Interfaces*. ACM, 2026, pp. 2004–2018.

[9] M. Yin, H. Zu, W. Gu *et al.*, "How contextualized generative AI shapes player experience in games," *Entertainment Computing*, vol. 58, p. 101194, 2026.

[10] V. Figueiredo and D. Elumeze, "Symbolically scaffolded play: Designing role-sensitive prompts for generative NPC dialogue," 2025, preprint.

[11] G. N. Yannakakis and J. Togelius, "Experience-driven procedural content generation," *IEEE Transactions on Affective Computing*, vol. 2, no. 3, pp. 147–161, 2011.

[12] D. Melhart, D. Gravina, and G. N. Yannakakis, "Moment-to-moment engagement prediction through the eyes of the observer: PUBG streaming on Twitch," in *FDG '20: Proceedings of the 15th International Conference on the Foundations of Digital Games*. ACM, 2020, pp. 1–10.

[13] Z. Wang, Q. Guo, R. Singh, and X. Hu, "Do vision language models understand human engagement in games?" 2026, preprint.

[14] L. Fan, G. Wang, Y. Jiang *et al.*, "MineDojo: Building open-ended embodied agents with internet-scale knowledge," in *Advances in Neural Information Processing Systems*, vol. 35. Neural Information Processing Systems Foundation, 2022, pp. 18 343–18 362.

[15] D. Pérez-Liébana, J. Liu, A. Khalifa, R. D. Gaina, J. Togelius, and S. M. Lucas, "General video game AI: A multitrack framework for evaluating agents, games, and content generation algorithms," *IEEE Transactions on Games*, vol. 11, no. 3, pp. 195–214, 2019.

[16] X. Liu, H. Yu, H. Zhang *et al.*, "AgentBench: Evaluating LLMs as agents," in *International Conference on Learning Representations*, 2024, pp. 52 989–53 046.

[17] C. Ma, J. Zhang, Z. Zhu *et al.*, "AgentBoard: An analytical evaluation board of multi-turn LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 74 325–74 362.

[18] X. Zhou, H. Zhu, L. Mathur *et al.*, "SOTOPIA: Interactive evaluation for social intelligence in language agents," in *International Conference on Learning Representations*, 2024, pp. 40 975–41 019.

[19] D. Paglieri, B. Cupiał, S. Coward *et al.*, "BALROG: Benchmarking agentic LLM and VLM reasoning on games," in *International Conference on Learning Representations*, 2025, pp. 96 666–96 702.

[20] P. Le Pelletier de Woillemont, R. Labory, and V. Corruble, "Automated play-testing through RL based human-like play-styles generation," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 18, no. 1, pp. 146–154, 2022.

[21] Y. Ma, J. Yan, B. Shi *et al.*, "Can LLM agents stick to the script? a benchmark for long-horizon consistency in interactive narratives," 2026, preprint (v1); ICML 2026 acceptance comment on arXiv, proceedings record unverified.

[22] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 10 932–10 952.

[23] L. Beurer-Kellner, M. Fischer, and M. Vechev, "Prompting is programming: A query language for large language models," *Proceedings of the ACM on Programming Languages*, vol. 7, no. PLDI, pp. 1946–1969, 2023.

[24] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 9895–9901.

[25] S. Kambhampati, K. Valmeekam, L. Guan *et al.*, "Position: LLMs can't plan, but can help planning in LLM-Modulo frameworks," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 22 895–22 907.

[26] B. Ichter, A. Brohan, Y. Chebotar *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of the 6th Conference on Robot Learning (CoRL 2022)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2023, pp. 287–318.

[27] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," 2025, preprint (v3); ICSE 2026 acceptance stated on arXiv, proceedings record unverified.

[28] Z. Zeng, S. Li, J. Xi, A. Zhu, and P. Ammanabrolu, "Setting the DC: Tool-grounded D&D simulations to test LLM agents," in *NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation (GenProCC)*, 2025, non-archival workshop paper; OpenReview/PDF list Zeng first, with Zeng and Li as equal contributors; NeurIPS/GenProCC lists display Li first.

[29] E. Zhou, S. Basavatia, M. Siam, Z. Chen, and M. O. Riedl, "STORY2GAME: Generating (almost) everything in an interactive fiction game," 2025, preprint.

[30] S. Buongiorno, L. Klinkert, Z. Zhuang, T. Chawla, and C. Clark, "PANGeA: Procedural artificial narrative using generative AI for turn-based, role-playing video games," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 20, no. 1, pp. 156–166, 2024.

[31] M. Vaucher, S. Silveira, S. Góngora, and L. Chiruzzo, "IVIE: A neuro-symbolic approach to incremental and validated generation of interactive fiction worlds," in *Proceedings of the 17th International Conference on Computational Creativity (ICCC'26)*. Association for Computational Creativity, 2026, archival proceedings (ISBN); preprint arXiv:2606.13348.

[32] S. Góngora, L. Chiruzzo, G. Méndez, and P. Gervás, "World-state transformations for neuro-symbolic interactive storytelling," 2026, preprint.

[33] S. Geng, H. Cooper, M. Moskal *et al.*, "JSONSchemaBench: A rigorous benchmark of structured outputs for language models," 2025, preprint (v3).

[34] X. He, Y. Tian, Y. Sun *et al.*, "G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 132 876–132 907.

[35] B. Jiménez Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 59 532–59 569.

[36] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *ECAI 2025*, ser. Frontiers in Artificial Intelligence and Applications, vol. 413. IOS Press, 2025, pp. 2993–3000.

[37] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 29 304–29 320.

[38] J. Pineau, P. Vincent-Lamarre, K. Sinha *et al.*, "Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.

[39] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," *Journal of Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.

[40] C. van der Lee, A. Gatt, E. van Miltenburg, S. Wubben, and E. Krahmer, "Best practices for the human evaluation of automatically generated text," in *Proceedings of the 12th International Conference on Natural Language Generation*. Association for Computational Linguistics, 2019, pp. 355–368.

[41] M. Karpinska, N. Akoury, and M. Iyyer, "The perils of using Mechanical Turk to evaluate open-ended text generation," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 1265–1285.

[42] L. Zheng, W.-L. Chiang, Y. Sheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems*, vol. 36. Neural Information Processing Systems Foundation, 2023, pp. 46 595–46 623.

[43] G. H. Chen, S. Chen, Z. Liu, F. Jiang, and B. Wang, "Humans or LLMs as the judge? a study on judgement bias," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 8301–8327.

[44] D. Lakens, A. M. Scheel, and P. M. Isager, "Equivalence testing for psychological research: A tutorial," *Advances in Methods and Practices in Psychological Science*, vol. 1, no. 2, pp. 259–269, 2018.

[45] D. J. Barr, R. Levy, C. Scheepers, and H. J. Tily, "Random effects structure for confirmatory hypothesis testing: Keep it maximal," *Journal of Memory and Language*, vol. 68, no. 3, pp. 255–278, 2013.

[46] E. Feng, W. Zhou, Z. Liu *et al.*, "Get experience from practice: LLM agents with record & replay," 2025, preprint.

[47] Y. Sun, H. Wang, D. Li, G. Wang, and H. Zhang, "The emperor's new clothes in benchmarking? a rigorous examination of mitigation strategies for LLM benchmark data contamination," 2025, preprint (v1).

[48] A. Madaan, N. Tandon, P. Gupta *et al.*, "Self-Refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 46 534–46 594.

[49] M. O. Riedl, C. J. Saretto, and R. M. Young, "Managing interaction between users and agents in a multi-agent storytelling environment," in *Proceedings of the Second International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2003, pp. 741–748.

[50] R. E. Fikes and N. J. Nilsson, "STRIPS: A new approach to the application of theorem proving to problem solving," *Artificial Intelligence*, vol. 2, no. 3–4, pp. 189–208, 1971.

[51] F. B. Schneider, "Enforceable security policies," *ACM Transactions on Information and System Security*, vol. 3, no. 1, pp. 30–50, 2000.

[52] T. Haerder and A. Reuter, "Principles of transaction-oriented database recovery," *ACM Computing Surveys*, vol. 15, no. 4, pp. 287–317, 1983.

[53] R. Evans and E. Short, "Versu—a simulationist storytelling system," *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 6, no. 2, pp. 113–130, 2014, predecessor title of IEEE Transactions on Games.

[54] S. G. Ware and C. Siler, "Sabre: A narrative planner supporting intention and deep theory of mind," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 17, no. 1, 2021, pp. 99–106.

[55] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 8634–8652.

[56] X. Chen, M. Lin, N. Schärli, and D. Zhou, "Teaching large language models to self-debug," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024, accepted poster; OpenReview archival record; preprint arXiv:2304.05128.

[57] A. Solar-Lezama, L. Tancau, R. Bodík, S. Seshia, and V. Saraswat, "Combinatorial sketching for finite programs," in *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. ACM, 2006, pp. 404–415.

[58] Y. Huang, G. Chen, J. Yao *et al.*, "Beyond state consistency: Behavior consistency in text-based world models," 2026, preprint.